Fuel Cell – Maintenance Practices

1. Removal/Installation

A. Remove Fuel Cell

1) Ensure all electrical power is disconnected, crossfeed valve is closed, and wing fuel tank switches (if fitted) are OFF.

2) Drain fuselage fuel tanks. (Refer to 12-10-10, Fuel System – General Data).

3) Remove appropriate access panels from underside of fuselage for cell being removed. (Refer to 12-00-00, Servicing – General Data).

4) Remove relevant sump plate. (Refer to 28-10-21, Fuel Cell Sump Plate – Maintenance Practices).

5) Working from inside fuel cell, except No. 4 and 5 cells, remove clamp from interconnect pipe molded fitting and pull cell from interconnect pipe.

6) If either No. 1 or No. 7 fuel cell is being removed, proceed as follows (refer to Figure 1):

a) Working from inside fuel cell, remove bolts and sealing washers securing cell to fuselage structure around filler neck.

b) Withdraw retaining ring, gasket, preformed packing, and retaining washer from fuel cell.

7) If No. 6 fuel cell is being removed, proceed as follows (refer to Figure 1):

a) Remove main landing gear fairing from each side of fuselage. (Refer to 12-00-00, Servicing – General Data).

b) Disconnect flexible hose from external end of each main feed line connector.

c) Remove nut, washer, bolt, and hose clamp securing each main feed line connector to support bracket on fuselage structure.

d) Working from inside fuel cell, remove bolts and sealing washers securing each main feed line connector and cell to fuselage structure.

e) Remove both main feed line connectors, with attached flexible hoses, from inside fuel cell.

8) Remove necessary heating ducts from both sides of cabin to provide access to vent system lines of fuel cell being removed.

9) Remove hose clamps securing molded vent outlets at each side of cell to left and right vent lines of fuel system. Pull cell vent outlets from fuel system vent lines.

10) Disengage tape strips along top, front and rear sides of fuel cell.

11) Ensure that fuel cell is warm enough to be flexible, fold cell carefully, and remove through access opening in lower surface of fuel cell compartment in fuselage structure.

B. Prepare Fuel Cell Compartment for Installation of Cell

1) Check that all fuselage structure rivet heads and tails, exposed inside fuel cell compartment, are covered with Scotch foam tape 3M4116 (Minnesota Mining and Manufacturing Co.).

2) Check that edges of all frame flanges, stringers and skin joints, exposed inside fuel cell compartment, are covered with Scotch foam tape 3M4116.

3) Check that Velcro tape strips are securely bonded to underside of sub-floor structure at all frame faces with adhesive compound EC1099 (Minnesota Mining and Manufacturing Co.).

4) Check that covers at frame stations 218.82, 239.88 and 262.00, in cell compartments 5 through 8, are securely bonded to the fuselage structure with adhesive compound EC1099, or with 3M–5952WF (or 3M–5952) double coated tape.

C. Install Fuel Cell

1) Inspect fuel cell compartment for cleanliness.

2) Observe following fuel cell handling precautions:

a) Do not remove fuel cell from shipping containers until required for installation and then only in immediate vicinity of fuel cell compartment.

b) Carefully examine fuel cell and container for possible damage caused during packing, transportation and unpacking.

c) Immediately after removal from shipping container, place fuel cell on clean, heavy paper.

d) Do not lift, carry or pull fuel cell by hangers or fittings.

e) Check that fuel cell is completely flexible before handling. Temperature should not be lower than 4.5°C (40°F) and preferably at room temperature, approximately 21°C (70°F).

f) Always use original container when moving or shipping fuel cells.

3) Insert fuel cell into appropriate compartment through access opening in lower surface of fuselage structure.

4) Carefully unfold fuel cell and engage Velcro tape strips along top, front and rear sides.

5) Using Petrolatum (Specification VV–P–236) or equivalent, as a lubricant, push molded vent outlet at each side of fuel cell over end of corresponding left and right vent line of fuel system. Secure joints with hose clamps.

6) Reinstall heating ducts, removed for access to vent lines, at each side of cabin.

7) Working from inside fuel cell, except No. 4 and No. 5 cells, push cell interconnect pipe molding over interconnect pipe and secure joint with hose clamp.

8) If No. 6 cell is being installed, proceed as follows (refer to Figure 1):

a) Insert left and right main feed line connectors, with attached flexible hoses, inside fuel cell.

b) Working from inside fuel cell, pass outer end of left main feed line connector through hole in cell and left side of fuselage structure. Repeat operation for right main feed line connector.

c) Align mounting flange on each main feed line connector and secure to structure with bolts and sealing washers. Torque bolts to 35 to 40 pound–inches.

d) Position hose clamp on end of each main feed line connector projecting from fuel cell and secure clamps to support brackets on fuselage structure with bolts, washers and nuts.

e) Connect engine supply flexible hoses to main feed line connectors.

f) Reinstall main landing gear fairings. (Refer to 12-00-00, Servicing – General Data).

9) If either No. 1 or No. 7 fuel cell is being installed, proceed as follows (refer to Figure 1):

a) Position opening in fuel cell over and around filler neck and align bolting holes in cell and fuselage structure.

b) Place retaining washer over filler neck with projecting collar of washer inserted between filler neck and holes in fuel cell and fuselage structure.

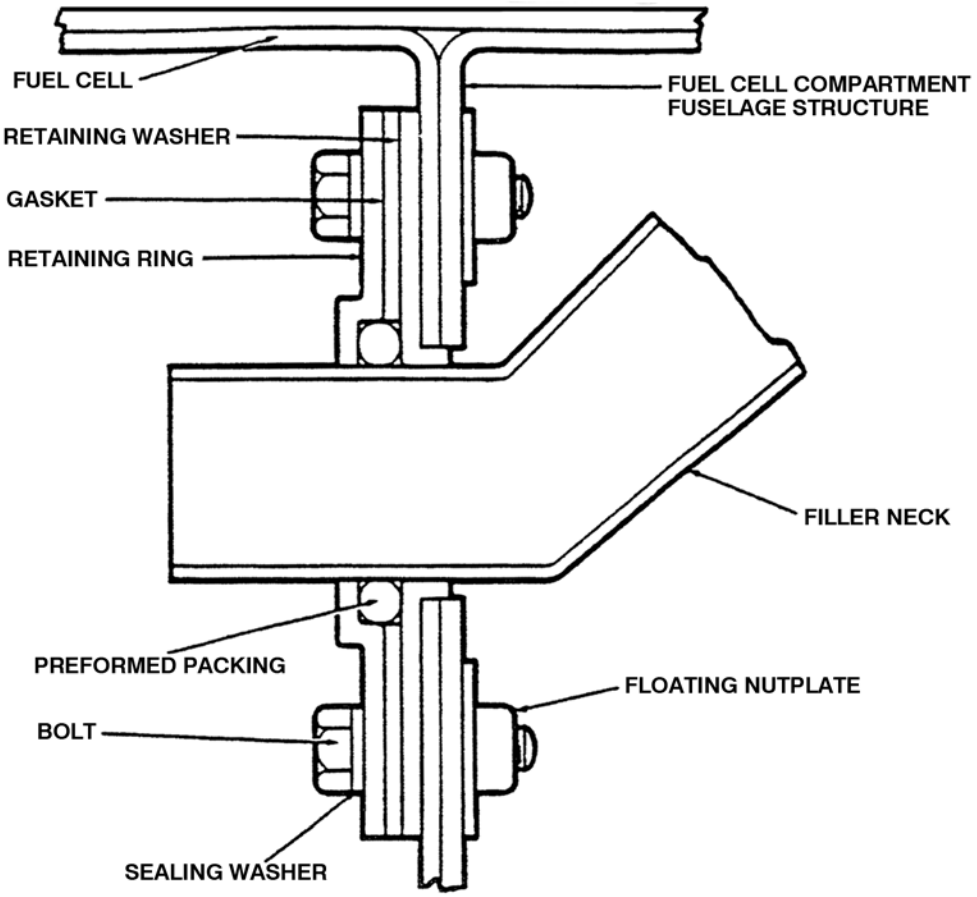
c) Place gasket and preformed packing over filler neck, followed by retaining ring. Ensure that dished center of retaining ring is positioned to enclose preformed packing.

d) Align bolting holes in all parts and secure with bolts and sealing washers. Torque bolts to 35 to 40 pound–inches.

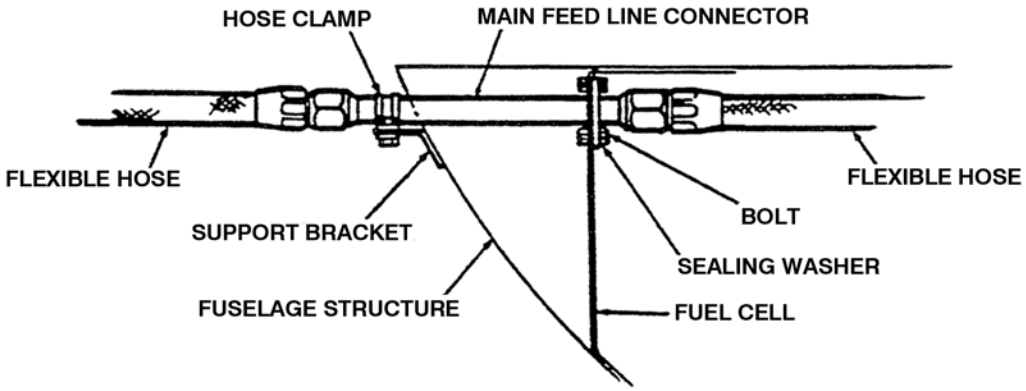
10) Install sump plate. (Refer to 28-10-21, Fuel Cell Sump Plate – Maintenance Practices).

11) Check installed fuel cell for leaks and on cell Nos. 4, 5 and 6, carry out appropriate checks as detailed in 28-00-00, Fuel (Mod 6/2006) – General Data, and 28-00-01, Fuel (With Mod S.O.O. 6247) – General Data.

12) Install access panels on underside of fuselage. (Refer to 12-00-00, Servicing – General Data).



CROSS SECTION THROUGH FILLER NECK
(TYPICAL FOR NO.1 AND NO.7 FUEL CELLS)



CROSS SECTION THROUGH NO.6 FUEL CELL
(TYPICAL BOTH SIDES)

Attachment of Filler Neck and Main Feed Lines in Fuel Cells
Figure 1